# DSBDA Practical 13

Input files:

## Step 2: Create Dataset File

nano sample_weather.txt

Paste the following data:

2026-03-01 10:00 30 22 15

2026-03-01 11:00 32 23 18

2026-03-01 12:00 35 24 20

2026-03-01 13:00 36 25 22

2026-03-01 14:00 37 26 25

2026-03-02 10:00 29 21 14

2026-03-02 11:00 31 22 17

2026-03-02 12:00 34 23 19

2026-03-02 13:00 35 24 21

2026-03-02 14:00 36 25 23

## Step 3: Create Mapper

nano mapper.py

```
#!/usr/bin/env python3
import sys
for line in sys.stdin:
parts = line.strip().split()
if len(parts) == 5:
print(f"TEMP\t{parts[2]}")
print(f"DEW\t{parts[3]}")
print(f"WIND\t{parts[4]}")
```

## Step 4: Create Reducer

nano reducer.py

```
#!/usr/bin/env python3
import sys
data = {}
```

for line in sys.stdin:

key, value = line.strip().split("\t")

value = float(value)

if key not in data:

data[key] = [value,1]

else:

data[key][0]+=value

data[key][1]+=1

for key in data:

total,count=data[key]

print(f"{key}_avg\t{total/count}")

## Step 5: Give Permissions

chmod +x mapper.py reducer.py

## Step 6: Run MapReduce Job

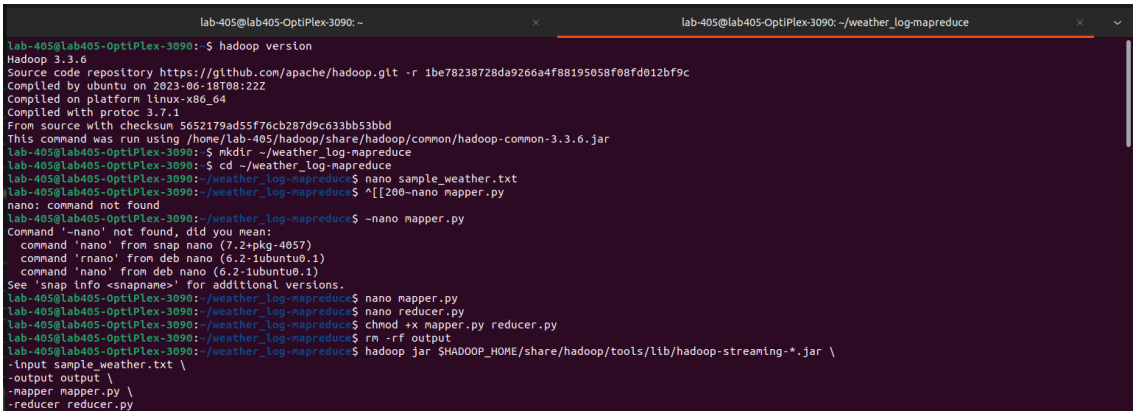
rm -rf output

hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \

- input sample_weather.txt \
- output output \
- mapper mapper.py \
- reducer reducer.py

## Step 7: View Output

cat output/part-00000



```
                Bytes Read=262
        File Output Format Counters
                Bytes Written=53
2026-04-16 12:30:17,191 INFO streaming.StreamJob: Output directory: output
lab-405@lab405-OptiPlex-3090:~/weather_log-mapreduce$ cat output/part-00000
DEW_avg 23.5
TEMP_avg        33.5
WIND_avg        19.4
lab-405@lab405-OptiPlex-3090:~/weather_log-mapreduce$
```

# 5. Conclusion

The MapReduce program successfully computes averages using distributed processing concepts.